

ChatGPT

Purpose: To size the main passive components of an LT1959 buck converter to generate 3.3 V / 200 mA from a rated 7.4 V 2S LiPo battery, with a maximum requested output ripple of 50 mV and an inductor current ripple of 30%.

1. Data and assumptions

Size	Value	Comment
Vin	7.4 V	Rated Voltage 2S LiPo Battery
Vout	3.3 V	Microprocessor power supply
Is	200 mA	Rated Current
ΔV_S max	50 mV peak-to-peak	Maximum Allowable Ripple at the STM32 Input
ΔI_L	30% Is = 60 mA i.e.	Inductor Current Ripple
fs	500 kHz	LT1959 Fixed Frequency
VFB	1.21 V	FB Pin Reference

Important note: the calculations below are made at the rated point $V_{in} = 7.4$ V. For a series design, it is also necessary to check the fully charged battery, typically 8.4 V for a 2S LiPo, and the dead battery. As the LT1959 is greatly oversized in current for 200 mA, the dominant constraints will be more ripple, stability, choice of output capacitor and routing.

2. Principle Electrical Diagram

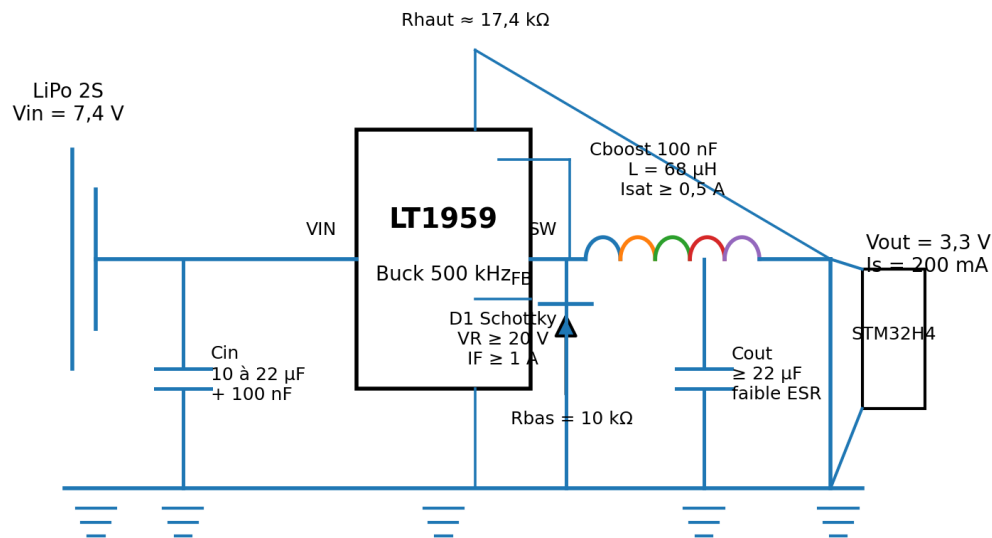


Schéma de principe pour le dimensionnement. Les valeurs finales doivent être validées avec la compensation, le routage et la fiche technique.

Figure 1 - Schematic diagram of the LT1959 to 3.3 V buck converter.

3. Duty cycle

As a first approximation, neglecting the drops in the internal switch and in the freewheel diode:

$$D = V_{out} / V_{in} = 3.3 / 7.4 = 0.446, \text{ i.e. } 44.6\%.$$

At 500 kHz, the switching period is $T = 1 / f_s = 2.00 \mu\text{s}$. We obtain $T_{\text{on}} = D \cdot T = 0.89 \mu\text{s}$ and $T_{\text{off}} = (1-D) \cdot T = 1.11 \mu\text{s}$.

4. Inductor Sizing

For a buck in continuous conduction mode, the peak-to-peak ripple of the inductor current is:

$$\Delta I_L = (V_{\text{in}} - V_{\text{out}}) \cdot D / (L \cdot f_s)$$

$$\text{So } L = (V_{\text{in}} - V_{\text{out}}) \cdot D / (f_s \cdot \Delta I_L) = (7.4 - 3.3) \times 0.446 / (500 \text{ kHz} \times 60 \text{ mA}) = 60.9 \mu\text{H}.$$

Recommended normalized value: 68 μH . With 68 μH , the ripple is about 53.8 mA p-to-p, or about 26.9% of the rated current.

Peak current in the inductor: $I_{\text{pk}} = I_s + \Delta I_L / 2 = 200 \text{ mA} + 30 \text{ mA} = 230 \text{ mA}$. Minimum current: $I_{\text{min}} = 170 \text{ mA}$. Choose an inductor with a much higher saturation current, e.g. $I_{\text{sat}} \geq 0.5 \text{ A}$, and a low series resistor to limit losses at low voltages.

5. Output capacitor

The output ripple has two main contributions: capacitive load/discharge and ESR drop of the capacitor. The ideal capacitive component is:

$$\Delta V_{\text{cap}} = \Delta I_L / (8 \cdot f_s \cdot C_{\text{out}})$$

$$\text{Imposing } \Delta V_{\text{cap}} \leq 50 \text{ mV: } C_{\text{out}} \geq \Delta I_L / (8 \cdot f_s \cdot \Delta V) = 60 \text{ mA} / (8 \times 500 \text{ kHz} \times 50 \text{ mV}) = 0.30 \mu\text{F}.$$

The purely capacitive stress is therefore very low. The ESR constraint gives: $\text{ESR} \leq \Delta V / \Delta I_L = 50 \text{ mV} / 60 \text{ mA} = 0.83 \Omega$.

Recommended practical choice: at least 22 μF ceramic X7R in the output, supplemented if necessary by 100 nF close to the load. The actual value of a ceramic capacitor decreases with DC polarization, so the effective capacitance at 3.3 V must be checked. For a microprocessor, the goal is not only ripple at 500 kHz, but also to withstand fast current demands.

6. Input capacitor

The input capacitor must provide the pulsed current from the converter and remain close to the VIN/GND pins of the LT1959. Its approximate RMS current is:

$$I_{\text{cin,rms}} \approx I_s \cdot \sqrt{D \cdot (1-D)} = 0.2 \times \sqrt{0.446 \times 0.554} = 99 \text{ mA RMS}.$$

To aim, for example, for 100 mV of low-frequency switching input ripple, an estimate gives $C_{\text{in}} \geq I_s \cdot D \cdot (1-D) / (f_s \cdot \Delta V_{\text{in}}) = 0.99 \mu\text{F}$. In practice, retain 10 to 22 μF low ESR, with a ceramic 100 nF directly as close as possible to VIN and GND.

7. FB Return Bridge

The output voltage is set by: $V_{\text{out}} = V_{\text{FB}} \times (1 + R_{\text{haut}}/R_{\text{bas}})$.

$$\text{By choosing } R_{\text{bas}} = 10 \text{ k}\Omega: R_{\text{haut}} = 10 \text{ k}\Omega \times (3.3/1.21 - 1) = 17.27 \text{ k}\Omega.$$

Normalized value: $R_{\text{haut}} = 17.4 \text{ k}\Omega$ and $R_{\text{bas}} = 10 \text{ k}\Omega$, which gives $V_{\text{out}} \approx 3.32 \text{ V}$. An alternative is $R_{\text{haut}} = 17.2 \text{ k}\Omega$ to obtain about 3.29 V.

8. Freewheel Diode and Boost Capacitor

The freewheel LED drives during Toff. Its average current is $I_{d,avg} \approx I_s \cdot (1-D) = 0.2 \times 0.554 = 111 \text{ mA}$. Its peak current is close to the inductance peak current, i.e. about 230 mA at the rated point.

Choose a fast Schottky diode with 20 V VRRM and $I_F \geq 1$ for a comfortable headroom. The reverse voltage must at least support V_{in} , and the routing must minimize the LT1959 loop - diode - input capacitor.

The boost capacitor used by the internal control is typically 100 nF, placed as close as possible to the BOOST and SW pins, according to the application diagram of the data sheet.

9. Expected Waveforms

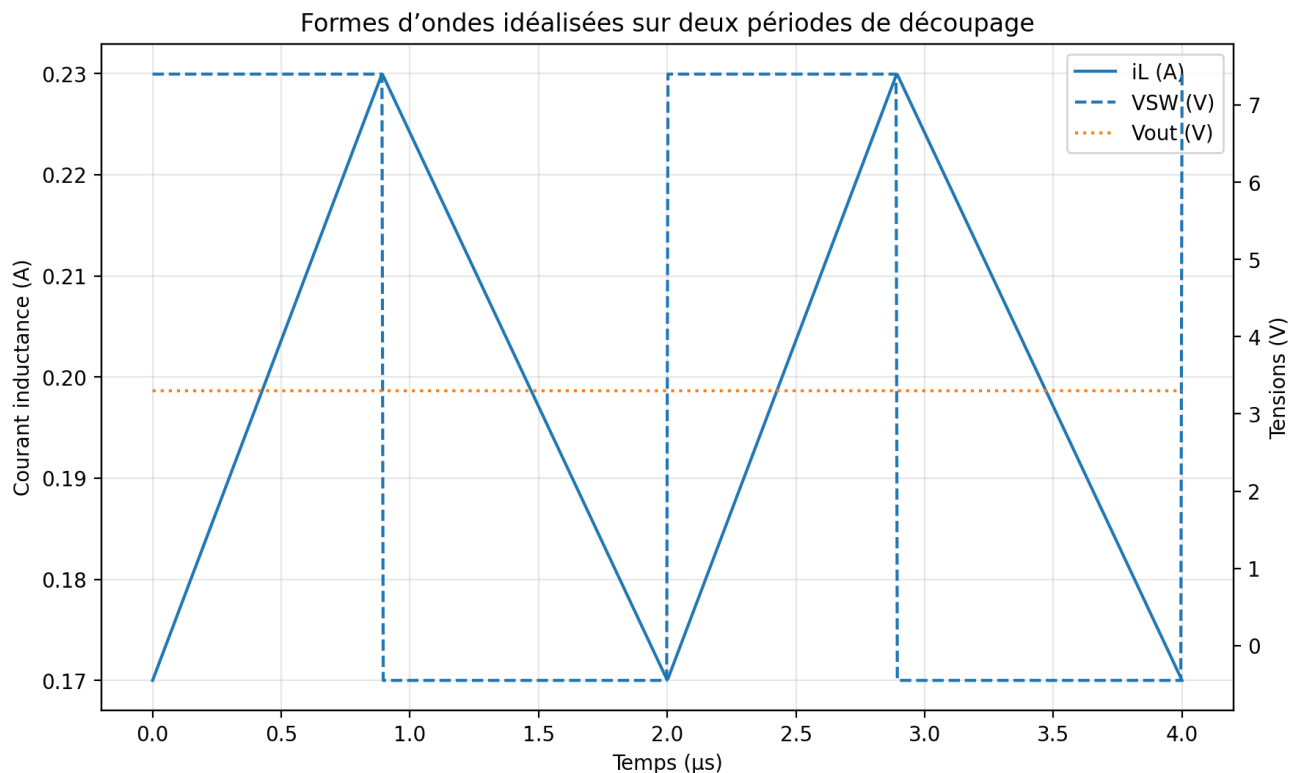


Figure 2 - Current in the inductor, SW voltage and output ripple idealized.

The inductance current is triangular in continuous conduction. The SW pin switches between a value close to V_{in} during Ton and a slightly negative voltage during Toff, limited by the freewheel diode. The output voltage is almost continuous, with residual ripple dependent on ΔI_L , Cout, ESR and routing.

10. Summary of the proposed values

Component	Calculated Value / Constraint	Proposed choice
L	60.9 μ H for $\Delta I_L = 60$ mA	68 μ H, $I_{sat} \geq 0.5$ A
Cout	≥ 0.30 μ F and $ESR \leq 0.83$ Ω	22 μ F X7R + 100 nF, low ESR
Cin	$I_{rms} \approx 99$ mA; Low minimum capacity	10 to 22 μ F low ESR + 100 nF near LT1959
Rhaut/Rbas	$R_{haut} = 17.27$ k Ω as $R_{bas} = 10$ k Ω	17.4 k Ω / 10 k Ω
D1	$I_{d,avg} \approx 111$ mA; Peak ≈ 230 mA	Schottky ≥ 20 V, ≥ 1 A
Cboost	According to LT1959 application	100 nF near BOOST-SW

Conclusion: At the rated point 7.4 V to 3.3 V / 200 mA, a 68 μ H inductance and a 22 μ F ceramic output capacitor give a comfortable margin with respect to the 50 mV ripple. However, loop stability must be validated with VC pin compensation, actual capacitor characteristics, and PCB routing.

References

Analog Devices, LT1959 product page and data sheet: 500 kHz buck regulator, 1.21 V FB adjustable output, 4.5 A switch, 4 V minimum input, VIN/GND/SW/diode routing recommendations.